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Comments:			
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gangopadhyay, Anup Ph.D. (Washington University in St. Louis)		
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	This grant is a continuation of previous grants NNX10AU19G and NNX16AB52G. Funding is provided for Prof. Dr. Kenneth Kelton and Dr. Anup Gangopadhyay for their role in incorporating all ground-based studies and flight preparation activities by the QUASI team at Washington University in St. Louis (WUSTL). These support QUASI flight investigations on the International Space Station (ISS) and collaborative flight projects with ESA teams ICOPROSOL (scientific coordinator – Dr. Dirk Holland-Moritz, DLR) and THERMOPROP (scientific coordinator – Prof. Dr. Hans Fecht, Ulm University). The flight experiments on the ISS are in designated batches. Batch 1 and batch 2 experiments have now been successfully completed. The planned activities under this grant are for samples approved for batches 3 and 4. Metallic alloy liquids processed in a containerless environment are investigated in all of the proposed measurements for		

Task Description:

QUASI, THERMOPROP and ICOPROSOL. The primary goals of the flight and ground-based experiments are:

1. An investigation of the role of stirring and diffusion on crystal nucleation and a comparison of the experimental results with predictions of a new model for nucleation, the Coupled Flux Model;

2. Measurements of key parameters needed for a quantitative analysis of the nucleation data obtained in the ground-based and ISS studies (driving free energy, specific heat and viscosity);

3. Measurements of the electrical resistivity of metallic liquids as a function of temperature to investigate a proposed correlation between structural changes in the liquid and dynamical properties, and to confirm the existence of a plateau in the temperature dependence of the resistivity that we have recently discovered at high temperatures;

4. Measurements of the thermal conductivity of metallic liquids at high temperature;

5. Measurements of the crystal growth velocity in the ground-based and ISS studies.

Rationale for the Research

Measurements of the maximum supercooling of free-cooling metallic liquids that crystallize to phases of a different composition than the liquid, are made in the quiescent environment of the electrostatic levitation facility (WU-BESL) located in our laboratory at Washington University in St. Louis. Long-range diffusion is required for nucleation in these cases. These ground-based results are compared with undercooling studies made as a function of stirring in the ISS-EML. According to the Coupled Flux Model for nucleation, developed by Kelton, the nucleation rate should increase with the stirring rate, since long-range diffusion will become less important. A key goal of our ongoing research is to investigate this.

Measurements of the thermophysical properties, including the specific heat, density, surface tension, viscosity, thermal conductivity and electrical resistivity, will be made using the WU-BESL and the ISS-EML, since these facilities have different capabilities for these studies. The specific heat, density, surface tension and viscosity are needed for a quantitative analysis of the nucleation data. Since our earlier studies have demonstrated that ordering in the liquid couples to the nucleation barrier, the undercooling and thermophysical property measurements will be correlated with x-ray synchrotron diffraction results for the supercooled liquids of interest, obtained previously using WU-BESL. Based on our previous studies, the specific heat and electrical resistivity are also valuable for providing structural evidence for a cooperative cross-over behavior in the shear viscosity at high temperatures. When coupled with the measured electrical resistivity, measurements of the thermal conductivity will allow an investigation of the Widemann-Franz law for high temperature metallic liquids, a case where it is not been studied before.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes, cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. The combination of the containerless based methods of WU-BESL and the ISS-EML enables these measurements. In addition to their relevance for processes such as additive manufacturing on earth, where nucleation in highly stirred conditions is important, measurements in two different gravitational fields will be important for designing additive manufacturing in space and on the Moon. Finally, past measurements of the electrical resistivity in high temperature liquids have revealed a plateau in the temperature dependence of the resistivity at temperatures near a crossover temperature for the viscosity. This is not predicted by any existing theory. Our measurements will provide valuable information for developing a theoretical understanding of this phenomenon.

A summary of activities carried out during year 3 of this grant are listed below.
1. Measured the heats of fusion for Zr80Pt20, Cu47Zr47Al6, and Ti45Zr45Ni10, which are necessary for the quantitative analysis of nucleation data obtained for these alloys in ground-based and International Space Station (ISS) experiments. The measured values are in conflict with the values estimated from the commonly used Richard's rule.

2. Measured the specific heats at constant pressure for Zr80Pt20, Cu47Zr47Al6, and Ti45Zr45Ni10. Several anomalies were noted, such as multiple maxima in the Zr80Pt20 liquid. Further experimental studies are planned to ensure that these are not experimental artifacts.

3. Crystal nucleation rates were measured in the quiescent environment of the ground-based electrostatic levitation (ESL) studies and on the ISS as a function of stirring levels in the liquid for Zr80Pt20, Cu47Zr47Al6, and Ti45Zr45Ni10 liquids. The key results are listed for each alloy:

a. Zr80Pt20 – According to the equilibrium phase diagram, this liquid should crystallize to a eutectic phase mixture of the Zr solid solution and the Zr5Pt3 intermetallic phase. Instead, our synchrotron X-ray diffraction studies indicate that the liquid first crystallizes to a metastable icosahedral phase (i-phase), which subsequently transforms to the eutectic phase mixture. Based on the Coupled Flux model for nucleation, the dynamical pre-factor in the nucleation rate should increase with stirring. In contrast to this expectation, it decreases with increased stirring in Zr80Pt2. Molecular dynamics studies of the liquid are underway to understand whether these features are due to significant icosahedral ordering in the liquid. Few studies exist of the nucleation step for eutectic solidification. Our results provide new insight into that process.

b. Cu47Zr47Al6 – A decrease in the nucleation pre-factor with stirring that is much smaller than in Zr80Pt20 is observed, although poor statistics make this conclusion uncertain. The wide distribution in the undercooling temperatures suggest that nucleation may be to competing phases. Synchrotron X-ray data are currently being analyzed to investigate this possibility.

Task Progress:	c. Ti45Zr45Ni10 – The statistics of the ground-based ESL data are very good; however, the statistics are limited in the ISS experiments. It seems clear, however, that unlike the Zr80Pt20 and Cu47Zr47Al6 liquids, the nucleation pre-factor increases with increased stirring, in agreement with the predictions of the Coupled Flux model. In summary, based on the data obtained thus far, stirring clearly has an impact on the nucleation behavior. However, the dynamical pre-factor increases with stirring in some alloy liquids, but it decreases in others. A goal for future studies is to identify the mechanisms that are responsible for this different behavior. 4. Planned studies in the next year are: a. Additional thermophysical property measurements on Zr80Pt20 to determine whether the maxima in the specific heat near but below the melting (liquidus) temperature is real or an experimental artifact. b. Planned experimental studies on Pd43Cu27Ni10Pt20 and Zr50Pd50 in Batch 4.2, which will most likely occur in 2025. c. Molecular dynamics investigations of the Zr80Pt20 liquid to assess the mechanism of metastable i-phase nucleation and in Cu47Zr47Al6 to determine whether nucleation is to competing crystal phases. d. Preparation and submission of a manuscript discussing nucleation as a function of stirring in Zr80Pt20. Manuscripts on thermophysical property measurements are also planned. 5. Anup Gangopadhyay, Co-PI on this grant, will support an in-person international working group (IWG) meeting that will be held May 21-22, 2024 at the German Aerospace Center / Deutsches Zentrum fur Luft- und Raumfahrt (DLR). These accomplishments and planned activities are described in greater detail in the annual technical document. [Ed. Note: a technical progress update is delivered annually to NASA.]
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Matson DM, Battezzati L, Galenko PK, Gandin CA, Gangopadhyay AK, Henein H, Kelton KF, Kolbe M, Valloton J, Vogel SC, Volkmann T. "Electromagnetic levitation containerless processing of metallic materials in microgravity; rapid solidification." npjMicrogravity. 2023 May 2;9(1):34. https://doi.org/10.1038/s41526-023-00281-4 , May-2023
Articles in Peer-reviewed Journals	Wang Z, Dmowski W, Wang H, Ashcraft R, Abernathy DL, Kelton KF, Egami T. "Real-space atomic dynamics in metallic liquids investigated by inelastic neutron scattering." Physical Review B. 2024 Jul 1;110(2):024309. https://doi.org/10.1103/PhysRevB.110.024309 , Jul-2024